

THE SHAPE OF THE TRAP

*(in-scene rewrite — companion draft,
staying inside her head)*

The moon had no name of its own
by the ninth month — a survey
designation, three letters and a
string of digits nobody said out
loud. Somewhere in the process of
pouring six thousand foundations
into ground that had never held
anything, the workforce started
calling it the Anvil instead, because
that was what it felt like to build on:
something meant to take a hammer
blow and hold its shape anyway.

Ilse heard it first from a shift supervisor reporting a schedule delay, said it back to herself twice, and kept it without asking anyone's permission. It was the first piece of language this project had produced that no Council chamber had drafted or redlined. It belonged to the people standing on the ground.

Corin's lab sat at the edge of the settlement, in a freight module that had been meant to hold three more months of prefab housing before Corin quietly took it for the power team instead. Ilse approved the requisition without argument. By month nine she understood which of the two problems actually threatened to end the project first.

The zero-point line died the way the Casimir line had died three weeks earlier — honestly, on the record, just slower, because it had more places to hide.

Kwame Osei ran the final tests himself, a physicist Corin had recruited specifically for his refusal to call a result promising until he'd personally tried to break it. He spent eleven days trying to make the rig fail in some way that would explain its refusal to scale, and on the twelfth day he brought Corin the answer nobody wanted, delivered the way a person delivers a diagnosis they've already grieved on

their own time so they can say it plainly to someone else.

"The vacuum fluctuation is real," Kwame said. "I want that on the record before anything else. It's not nonsense. It's measurable, and it will never be enough. The energy density climbs with surface area, and the geometry that would give us enough surface area to matter is larger than the moon we're standing on. This isn't an engineering problem we haven't solved yet. It's a ceiling built into the mechanism itself."

"How confident," Corin asked.

"As confident as I've ever been about anything I spent eleven days

trying to disprove and failed to disprove."

Ilse had watched an earlier run of the same test, behind the same containment window where she'd once watched the Casimir rig give up its own last number. The readout climbed for eleven seconds to a figure that made Corin lean forward, then settled back, inevitably, to the same fraction of what the project needed.

"That's the best this geometry does," Kwame said, watching it fall.

"It would run a research station indefinitely. It will never run what we're actually trying to build."

Corin closed the file that afternoon — third dead end in as many months — and didn't dress up the closing for the team that gathered after. "We are now formally out of approaches with existing research behind them," she said, in front of a whiteboard with an increasingly short list of what hadn't worked and an empty column, still, for what would. "It does not mean the problem is unsolvable. It means the solution isn't sitting in an archive waiting to be rediscovered. It has to be something nobody's thought to try yet, and I don't currently know what that is."

Ilse wasn't in that meeting. She was three kilometers away that same week, watching Sefu rewrite a staffing model for the fourth time in as many months, because the workforce had crossed six thousand with no sign of leveling before the ten thousand year-two projections called for.

She found Corin on the platform that evening.

"Kwame closed the zero-point line," Ilse said. Not a question.

"Cleanly. Honestly." Corin looked out over the Anvil's spreading lights — six thousand now, a number that would have seemed impossible from this platform nine months

earlier. "I keep thinking about something you said the night of the vote. That the mathematics was never the hardest part. That you didn't find the real baseline until you stopped asking the question you'd been asking for three years."

"You think we're asking the wrong question."

"I think every dead line on that board shares one assumption nobody's examined. We keep asking how to *extract* more energy. What if the constraint isn't extraction efficiency at all. What if we've been trying to squeeze water from a stone, when what we need was never a stone to begin with."

Ilse looked at her fully, and recognized the stillness — the same one she'd watched Corin hold once before, in a mess hall three years and a lifetime ago, right before she found the third hole that wasn't a hole.

"Say the rest of it."

"I don't have the rest of it yet. Give me a week."

He needed four days.

"Every approach we've tried treats energy as something you take from a system that would rather keep it," Corin told the full team on the fifth morning, the board wiped clean for

the first time in three months. "I want to ask a different question. What if there's a structure that doesn't resist extraction at all. Not something you take from. Something that only ever does one thing. Emit."

Noor Hadad said the word before anyone else could. She'd spent eleven years, before this project, on the negative results of exactly this search — an experience she mentioned to almost nobody, and had come here half hoping someone would eventually need those eleven years of failure.

"You're talking about a white hole."

"I'm talking about a white hole."

The room went quiet the way the mess hall had gone quiet years earlier — an idea large enough that nobody wanted to be first to either endorse or dismiss it.

Noor broke the silence, not with a technical objection but the plainer one underneath it. "This has never existed in nature. Not once, anywhere we've ever looked. I want us to at least consider that nature doesn't produce them because something about their existence is unstable in a way our mathematics hasn't caught."

"That's fair," Corin said. "I don't have a clean answer for it, the same way Ilse didn't have one for Iyoro in the

hearing. The mathematics permits it. I can't tell you nature's silence is meaningless. I think the only way to actually answer it is the way this project has answered every unanswerable question so far — build the smallest, most contained version of the test we can survive being wrong about, and let the physics tell us instead of the whiteboard."

"Walk us through it," Ilse said, because she'd learned, over three years of asking Corin exactly this, that the walking-through was where an idea either survived or fell apart.

"A black hole only accretes," Corin said. "A white hole is the same

mathematics, run in reverse. Pure outward flux. Nothing crosses the horizon inward, it only—"

"Nothing has ever crossed a horizon that direction," Noor cut in. "Not once. You're describing the reverse of something we've actually observed, which isn't the same as describing something that can exist."

"I know the difference," Corin said. "There's a research program right now, real, currently unsuccessful, hunting microscopic black holes at particle colliders. Extra-dimension theories, a lowered Planck scale, quantum gravity finally in reach of

collision energies we can actually build."

"Hunting for black holes," Noor said.

"Not the other thing."

"That's my point. Nobody's ever aimed that same architecture at the other solution to the same equation, because everyone chasing it wants the interesting half of the pair. The half that accretes. Nobody wants the half that only emits."

"Because a black hole is a discovery," Noor said, "and the other half is a mathematical curiosity nobody's ever needed."

"But you want it for exactly that reason," Ilse said.

"A source, not a sink," Corin said. "It doesn't fight you for the energy, because there's no extraction happening at all. If we can produce one, even microscopic, even briefly, we don't have a harvesting problem anymore. We have a containment problem. That's a category I actually believe we know how to attempt."

Containment ate the next six weeks, and Ilse found herself pulled in more directly than she'd been pulled into anything since the original theory, because she could not stay out of a room where the machine she'd proposed three years earlier was finally being drawn instead of gestured at.

"The inner boundary holds the horizon in place," Corin said, first working session. "Difficult engineering. Not new physics. The outer boundary is the actual invention, and I don't yet know what it has to be made of. It has to be grown as one continuous piece, no seams, because seams are where a structure this stressed eventually fails."

Noor brought the harvesting half the same session — emitter crystals seeded through the lattice, driven by SASER arrays, phonon lasers, real proven hardware. Run both channels together, tuned right, and the photons and lattice vibrations stop behaving as

separate things. A phonon-polariton condensate. Proposed, recently, as a real possibility. Never built.

Nobody had ever needed to harvest an emission this intense, this continuously, off a source this small.

"You're telling me the harvesting mechanism and the containment structure are the same object," Ilse said slowly.

"I'm telling you they have to be," Corin said. "The lattice isn't a container with a power system bolted on. It's one continuous structure doing both jobs, because any seam between containment

and harvesting is a seam under this much stress."

"What happens to one emitter crystal if it fails independently," Ilse asked. "Before the whole structure. Does the rest route around it, the way a fault in a distributed system usually gets absorbed?"

Noor took longer to answer than Ilse expected. "In an ordinary distributed system, yes. A condensate isn't ordinary. It's one coherent quantum state, not a collection of parts. If enough of it is coherent, a local failure doesn't stay local. I want to answer honestly rather than reassuringly — I don't know how many crystals can fail

before the condensate itself collapses, and I don't think we'll know until we've built something close enough to test."

"So the redundancy everyone in this room built a career assuming a large structure should have," Ilse said, "doesn't apply here at all."

"Not the way any of us were trained to expect," Corin said. "That's the honest cost of a mechanism this elegant. It doesn't have separate failure domains to isolate a problem inside of. It has one domain, the whole thing, all the time. I say that because I think everyone who works on this deserves to know what kind of object they're actually

building from day one, instead of discovering it only once something's gone wrong."

Corin said the rest of it plainly, in the same session, because the charter's honesty clause had never been written for the Council alone. "If this fails, it doesn't fail in one subsystem while the others hold. There is no subsystem. When it fails, it fails everywhere at once, because everywhere is the same object under the same stress. I don't currently know how you shut something like that down safely. I suspect the answer isn't a switch. I suspect it's a sequence, in a fixed order, over some span of time long enough to matter, and I don't have

that sequence, because we haven't built the thing it would apply to yet. I want that on the record now, before any of us gets far enough in to forget how early this still is."

Ilse thought of the Anvil — and, for the first time since she'd filed it unsuitable, of the other candidate world's rough housing sketch, the school penciled into its northeast corner, *room to grow* in handwriting that wasn't hers. She'd rejected that world because a family was already imagining a house on it. She understood now, hearing the condensate's one failure domain described out loud, that she'd been protecting against exactly this shape of catastrophe before she'd

had the physics to describe it. She didn't feel the earlier caution had gone too far. She felt it hadn't gone far enough.

"We build the containment shell somewhere the eventual shutdown sequence gets a real budget of time and isolation," she said. "Not bolted on once the physics is finished. Part of the design, from the geometry up, on the assumption that someday somebody's going to need to walk this back safely, and I want that person to have been thought about by us, now, before we've built anything they'll actually have to walk back."

Perrin found her an hour later, tablet in hand — its own reliable warning by now.

"The oversight committee wants a substantive briefing," Perrin said.

"Not workforce updates. What the power architecture actually *is*. Nine months of curious is turning into nine months of unanswered, and those aren't the same thing to a committee that approved a resource figure this size."

"Tell them what we told the engineering team an hour ago," Ilse said. "We have a candidate mechanism. It's theoretical, unbuilt, and the person who designed it just told a room full of physicists he

doesn't yet know how to shut it down safely if it ever needs shutting down. I'm not going to soften that for a committee any more than I'd soften it for myself."

"They're going to hear a project that just discovered its own machine might not have a safe off switch," Perrin said, "delivered by the person legally accountable for everything on this moon. Some of them will hear honesty. Some will hear confirmation of exactly what Marrow warned this chamber about."

"Then that's what they hear. We agreed to this cost before we agreed to anything else."

It didn't stay contained to the committee. Within a week, *without a safe off switch* had become settlement-floor shorthand, repeated in mess halls with varying accuracy, and Sefu brought it to her directly rather than let it keep spreading uncorrected.

"People are asking whether they're building their own coffin," Sefu said, flat, the way she delivered anything she judged too important to soften. "Not all of them. Most still believe in what this is trying to do. But belief and fear aren't opposites — I've watched both live in the same person for fifteen years of running sites like this. Your name is on every legal document this project

has. That means when something frightening leaks out of a closed meeting, your name is the one people attach the fear to."

"What do you want me to do. Tell them it isn't dangerous?"

"I'm not asking you to lie. I'm asking you to say the true thing to their faces instead of letting them hear a fragment of it secondhand, three shifts removed from the actual meeting. You've been careful about honesty. I think you've been careless, without meaning to, about silence."

She held a town session three days later, unscripted, and answered every question the way she'd once

answered Marrow, without softening what she didn't know into something more comfortable.

Ingrid Bakke asked the last question of the night, the same fitter from the training bay.

"You told us we'd know the material well enough to trust our hands to it once the geometry was finalized. Is it finalized now? Is what you told this room tonight the actual shape we're building toward, or is there more you don't have yet?"

"There's more I don't have yet," Ilse said, because it was true. "The mechanism is real, on paper, in a way it wasn't a month ago. The engineering underneath it isn't

finished. I don't know exactly when it will be. I'll tell you as soon as I do."

Ingrid nodded once and sat back down among neighbors she hadn't known a year earlier, and didn't ask a follow-up.

Corin's open question — what the lattice actually had to be grown from — took two more weeks, and the answer came from Kwame, drafting the first formal engineering spec, who hit a wall the physics alone had never made him examine.

"We can describe the geometry," he told the team. "We can describe the growth process in principle. What we can't do is actually grow it. The

material doesn't exist at the scale we need, as far as I can determine, anywhere humanity has looked."

"What material," Ilse asked.

"Lonsdaleite. Hexagonal diamond. Same carbon as ordinary diamond, arranged differently — stronger, more thermally resilient, which is exactly the profile this lattice needs. It's only ever been found in meteorite-impact fragments, graphite locked into diamond's arrangement by a single violent instant, never a lab process. There's been one successful synthesis attempt in the last year. It tops out at millimeter scale. Our lattice needs a structure large enough to

hold a horizon, grown from a material nobody's ever produced in a quantity larger than a fingernail."

"So we're stopped again," Noor said, not an accusation. A fact on the table.

"Not entirely." Something in Kwame's voice made Ilse look up.

"This project has one confirmed sample already. A trace fragment, recovered years ago from a comet that passed through this system during a routine survey, catalogued as an anomaly nobody followed up on, because nobody had reason to care what a comet was carrying. It's not enough to build anything. It's enough to know exactly what we're

looking for, down to the crystal structure. And it's enough, if the survey records are precise enough, to run the comet's trajectory backward."

The room went quiet in a different register than the white-hole silence — closer to the hearing chamber, the moment before a chain of consequence becomes visible to everyone at once.

"You're saying there might be more of it out past this system," Ilse said.

"I'm saying there's one lead, and it's cold, years old, and nobody's ever had a reason to chase it before now."

He brought the fragment to the next session, sealed in a case small enough to fit in one hand — a chip barely larger than a grain of sand that didn't look, to Ilse's eye, like anything worth an expedition. "I've run this confirmation four times since it was catalogued," Kwame said. "Not because I doubted it. Because I kept expecting to find the reason it couldn't be real, the way every other approach in this project has had one, and I never found it. It came from somewhere that isn't here. That's the entire extent of what I can tell you, and it's more than this project had a week ago."

Ilse held the case a long moment before passing it on, turning it to

catch the light against a fragment too small to see clearly without magnification. This was the opposite kind of impossible from the resource figure that had settled onto her name the night of the vote — that had been too large to hold in her head as anything but abstraction. This was small enough to close a fist around, and the whole mechanism Corin had spent five days discovering now depended on it completely.

She found Tavik on the platform that same evening, where he'd taken, over recent months, to spending his evenings alone.

"I have something for you," she said.

"Not an assignment yet. A trajectory, and a reason someone's eventually going to need to chase it."

Tavik turned from the rail, and something in his posture changed.

"Tell me."

She told him — the comet, the fragment, the cold trajectory nobody had ever had reason to follow — and watched him listen the way he'd listened to her original proposal years earlier, all the way through, without interrupting.

"You said this wasn't an assignment yet," he said, when she'd finished.

"It isn't. We don't have a ship. We don't have a survey command. We don't know if what's out there is findable at all, or whether the trajectory decays into noise a few systems out. I'm telling you because you asked me to remember, months ago, and because you're the only person on this project who's spent a career doing exactly what this is going to require."

Tavik looked out over the Anvil's spreading lights, thousands now, arranged across ground that had held nothing a year earlier.

"I've spent my whole life arriving after the thing already happened,"

he said finally. "Verdent. The evacuation. Forty years of grief I never got to spend usefully on anything except remembering. I would like, before I'm done, to be the one who goes and finds something, instead of the one who cleans up after it's already been lost." He looked at her directly. "Tell me when there's a ship. I'll be ready before it is."

"It won't be safe," Ilse said. "I've spent this whole year learning how much I don't know how to promise people, and I'd rather say that to you now than let you volunteer for something you think I've already made safe."

"I buried a hundred and forty thousand people who trusted a model that told them the ground under their feet was safer than it was," Tavik said. "I am not asking you to promise me safety. I'm asking you to let me go do something that might actually matter, on the far side of a risk I've chosen with my eyes open, instead of spending whatever years I have left watching other people build the thing I only ever got to argue for in a chamber."

Ilse didn't sleep well that night, though the reason had shifted from the shape it took on the night of the vote. She was no longer wondering

whether the theory would hold.

What kept her awake was smaller, and harder to hold: they had solved the *shape* of the machine. They had not solved how to build it, and the gap between those two things was an expedition into territory nobody had ever charted, chasing a trail years cold, toward an answer that might not exist at the end of it at all.

Corin found her still awake near the hour the moon's cycle called morning, and sat down without needing to ask what kept her from sleeping.

"Noor asked me something after the session broke up," Corin said.

"Whether I'd thought about the fact

that we're proposing to build the one object in this mechanism that's never once occurred in nature, anywhere, in the whole recorded history of observation. I told her I had. I don't think I told her the rest of what I actually think — that I find it almost comforting rather than frightening, and I don't know if that says something good about me or something I should be worried about."

"Say the rest of it," Ilse said, the way she always did.

"Nature never needed a white hole. It already had everything it needed to run for however many billion years — stars, gravity, the ordinary

machinery, doing what it does without anyone asking it to give anything back on purpose. We need one because we're asking for something nature never had to solve. I don't think the absence tells us the mathematics is wrong. I think it tells us we're the first thing in this universe, as far as anyone's ever observed, that actually needed one to exist."

"Careful with that word," Ilse said.

"Needed. We want it badly enough to try building it. That's not the same claim as the universe needing it to exist, and I don't think you actually believe those are the same thing either."

"No," Corin admitted. "I don't. What I believe is smaller and, I think, still true — nothing before us ever had a reason to reach for this particular impossible thing. That's not the universe needing us. That's just us being the first ones with a reason."

"Which do you think it is.
Frightening, or plain."

"Both," Corin said. "The same way the lever in the mess hall was both the most elegant idea I'd ever heard and the most expensive way anyone's ever found to be wrong about something. I've stopped expecting one reading to cancel the other out. I think we just keep carrying both, the whole way

through, however far through turns out to be."

They sat together a while longer, watching the same pale unmoving curve of the gas giant that had watched over every hard conversation this project had ever had, in a system that had never once, across its whole long silent history, needed to produce the thing two exhausted engineers were now quietly asking it, without permission, to finally hold.

Chapter three, straight through – Kwame's two closed lines, Corin's white-hole pivot, the condensate design and the no-redundancy cost,

*the leaked "no safe off switch" and
the town hall, the lonsdaleite
fragment, and Tavik getting his
expedition. Same checks running the
whole way: no line telling you what a
silence or a held look already told
you. Say the word for chapter four.*